

SummarAI

Individual Reflection

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Individual Reflection — Antonio

My Contributions

Whisper Optimization

I profiled the full pipeline to find where time was being spent. The assumption going in was that the LLM call would be the slow part — in practice, Claude Haiku returns in a few seconds, while Whisper on a 20-minute audio file was taking close to 8 minutes with the default configuration. I implemented four optimizations, all of them zero-cost:

Switched from the multilingual base model to base.en: The English-only variant is faster and more accurate for English audio.

Set beam_size=1 (greedy decoding): Removes beam search overhead (~2× faster) with negligible accuracy loss for our task.

Set cpu_threads to use all available cores: Transcription runs fully parallel.

Passed language='en' explicitly: Skips the automatic language-detection pass.

Result: Processing time dropped from 7 min 50 s to 2 min 12 s on the same 20-minute recording — a ~3.5× speedup with near-identical transcript quality.

Evaluation Script

I built data/eval/evaluate.py. The script loads all transcripts from eval_set.jsonl, calls the same analyze() function used in production, compares predicted action items against gold labels using the team's agreed matching rule, and outputs per-meeting results plus micro-averaged precision, recall, and F1 to data/eval/results.json. The matching logic uses normalized text similarity with a threshold of 0.6 and greedy one-to-one assignment so each gold item is matched at most once.

Installation Guide

I wrote docs/installation_guide.md covering Python version requirements, virtual environment setup, ffmpeg installation on macOS, Linux, and Windows, .env configuration, and the exact commands to run the server.

Labeled Transcripts

I contributed approximately 8 labeled transcripts to eval_set.jsonl, including sprint planning sessions and one deliberately ambiguous case where responsibility is implied but no specific person is named.

What I Learned

Working on this project gave me a much more concrete understanding of how ASR models like Whisper actually work under the hood — specifically the trade-off between multilingual and language-specific models, and what beam search is doing in a seq2seq decoder. Before this I understood beam search as a concept from the course; seeing it actually cost us 3.5× in latency with no measurable quality benefit made the accuracy-efficiency trade-off feel real rather than theoretical. I also developed a much clearer

sense of what precision and recall mean as evaluation metrics for an information extraction task. A single accuracy number would have hidden the difference between the model missing real action items (recall failures) and inventing ones that were not there (precision failures) — two completely different failure modes with different implications for trust.